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Sub : Physics Assignment 4

Physics Assignment -

Quantum Computing - Total 10 Marks

Q.1. What is Quantum Computing?

Quantum Computing is a type of computing that uses qubits instead of classical bits.

Superposition: A qubit can be 0 and 1 at the same time.

Entanglement: Qubits can be strongly correlated with each other.

Quantum Interference - helps increase the probability of correct results.

Applications: Cryptography, optimization, drug discovery, AI and Simulations.

Q.2. Deutsch-Jozsa Algorithm.

The "Deutsch-Jozsa algorithm" determines whether a given function is constant or balanced.

Working:

1. Initialize qubits and apply Hadamard gates.
2. Pass them through the Oracle.

3. Pass them through the Oracle.

4. Apply Hadamard gates, qubits and measure measurement. $0 \rightarrow$ Constant, $1 \rightarrow$ Balanced.

Significance: It demonstrates that quantum computers can solve certain problems faster than classical computers.

Q.3. Schrodinger's Cat theory & Smully theory

Schrodinger's Cat illustrates Superposition a cat is theoretically considered both alive and dead until it is observed.

Quantum example - a qubit can exist as

Smully theory - the smully concept is used analogy for quantum superpositions and measurement. Smully can represent different possible states until observation determine one state.

Example a quantum state may have multiple possibilities, but measurement gives only one definite result.

Q.4. BB84 protocol?

BB84 is a quantum key distribution protocol proposed by Bennett and Brassard in 1984. It securely generates a shared secret key.

Steps:

1. Alice sends photons using randomly chosen bases.
2. Bob randomly chooses bases to measure them.
3. They publicly compare bases and keep matching results.

→ Security - any attempt to measure the quantum states changes them, causing detectable errors.

